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Long-Term Effect Estimation When Combining Clinical Trial and Observational Follow-Up Datasets

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ABSTRACT

Combining experimental and observational follow-up datasets has received much attention lately. In a survival setting, recent work has used Medicare claims to extend the follow-up period for participants in a prostate cancer clinical trial. This allows the estimation of the long-term effect that cannot be estimated by the trial data alone. In this article, we study the estimation of long-term effect when participants in a clinical trial are linked to an observational follow-up dataset. Such linkages are often incomplete and we formulate incomplete linkages as a missing data problem. We use the popular Cox model to define the long-term effect and we propose two approaches to deal with the missing data problem. The first approach, termed non-linked-as-censored (NLAC), is a simple approach that works when Cox model is correctly specified and linkage satisfies a conditionally independent assumption. To gain robustness against model mis-specification, we propose an inverse probability of linkage weighted approach, along with the augmented inverse probability of weighted method, based on a novel conditional linking at random (CLAR) assumption. We further extend our approach to incorporate time-dependent covariates. Simulation results confirm the validity of our method and we apply our methods to the SWOG study. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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Cox model; Incomplete linkage; Inverse probability weighting; Non linked as censored; Time-dependent covariate

1. Introduction

With the increasing availability of electronic health data, combining experimental and observational datasets has been widely applied in public health research (Warren et al. 2002; Gilbert et al. 2018). In a time-to-event setting, we consider the setup when data from a clinical trial is combined with an observational follow-up dataset, such as electronic health records or administrative claims. Clinical trials often study the effect of a particular treatment for a fixed period of time and it might not be long enough to determine the maximum benefit of the treatment. In contrast, an observational dataset such as Medicare claims naturally extends the follow-up period for clinical trial participants at minimal cost. This enables the estimation of the long-term effect for the treatment after the clinical trial. To combine the observational follow-up dataset with the clinical trial data, records belonging to the same individual can be linked with unique identifiers from both datasets. In this article, we use Cox model (Cox 1972) to define the long-term effect as the parameter for treatment when participants from the clinical trial are linked to an observational follow-up dataset.

For a real data example, the Prostate Cancer Prevention Trial (PCPT) was previously launched to examine whether finasteride¹ could prevent the development of prostate cancer (PC). PCPT showed that seven years of finasteride reduced PC risk by 25% (Thompson et al. 2003). However, it was unclear if seven years' of trial follow-up sufficed to determine the maximum

benefit of the treatment. Further, the reduced risk of PC for subjects receiving finasteride might not be maintained after finasteride discontinuation (Unger et al. 2018). A later study linked Medicare claims to the clinical records for participants in PCPT with their social security numbers (SSN) (Unger et al. 2018) to estimate the long-term effect of finasteride on PC development. In this example, PCPT is the clinical trial and Medicare claim is the observational follow-up dataset. Medicare claims extend the follow-up periods up to a maximum of 20 years compared to 7 years by the PCPT. Thus, we can observe more diagnosis times of PC within the Medicare claims dataset.

However, not every participant in the clinical trial can be linked to the observational dataset. For the PCPT-Medicare claims example, some participants might not be willing to share their SSNs or they may be enrolled in health maintenance organization (HMO) and Medicare claims are not applicable to HMO individuals (Unger et al. 2018). With incomplete linkages, survival outcomes in the observational dataset might be missing for some participants. For a participant censored in the clinical trial, meaning he was not diagnosed with PC within the clinical trial, his survival outcome in the observational dataset would be missing if he is unlinked. On the other hand, if a participant was diagnosed with PC within the clinical trial, his survival outcome has been already observed within the clinical trial and the linkage to the observational follow-up dataset is in fact not necessary. This suggests that the missingness of survival

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¹A treatment that inhibits the development potent androgen that fuels the malignancy of prostate cancer.

outcome depends both on the linkage status and whether a participant was censored in the clinical trial or not.

To address the missing survival outcomes, a complete-case analysis that only includes linked participants would ignore all the unlinked participants with observed survival outcomes within the clinical trial. However, simply adding those unlinked participants to the complete-case analysis also results in a biased estimate. This would cause the missingness of the survival outcomes to depend on itself and the missingness is then missing not at random (MNAR). To properly incorporate those unlinked participants with observed survival outcomes, we propose two approaches. The first approach, termed non-linked-as-censored (NLAC), treats unlinked participants who are censored in the clinical trial as censored, effectively ignoring their survival outcomes in the observational dataset. NLAC is a simple idea that works when the Cox model is correctly specified and the linkage satisfies certain conditionally independent assumption. However, when Cox model is mis-specified, we show that NLAC no longer provides unbiased estimates for the population parameters. To gain more robustness against model mis-specification, we develop a second approach based on an inverse probability of linkage weighted (IPLW) partial likelihood estimator. This approach involves modeling the linkage probability directly for unlinked participants that are censored in the clinical trial. We show that unbiased estimates can be obtained even if Cox model is mis-specified, under a novel assumption, termed conditional linking at random (CLAR). We further prove the asymptotic normality and consistency of the IPLW estimator. Our approach allows inclusion of time-dependent covariates for more flexibility. While there has been plenty work (Binder 1992; Robins 1993; Lin 2000; Qi, Wang, and Prentice 2005) on proving the asymptotic convergence for an inverse probability weighted (IPW) partial likelihood estimator when there are only time-independent covariates, their proof cannot be easily generalized to the case when there are time-dependent covariates (Breslow and Wellner 2007). To this end, we establish an IPW type empirical process weak convergence result that builds on the work in Saegusa and Wellner (2013) and borrow the techniques from Lin and Wei (1989) to extend the theoretical results to include time-dependent covariates. Finally, we develop an augmented inverse probability of linkage weighted (AIPLW) estimator. The estimator is doubly robust in the sense that it provides consistent estimates when either the linkage probability is consistently estimated or the involved regression functions are consistently estimated.

Related work. Our work contributes to the expanding body of literature on causal inference methods that integrate data from randomized clinical trials and observational studies, which leverage the strength from each for better power and external validity, see Colnet et al. (2024) for a comprehensive review. Indeed, there is a growing interest in combining different datasets and studying the treatment effect on long-term outcome in causal inference (Rosenman et al. 2018, 2020; Kallus and Mao 2020; Athey, Chetty, and Imbens 2020). All these work focuses on using experimental and observational datasets that contain different set of individuals, which is different from our setup. IPW has also been widely applied in the survival analysis setting (Binder 1992; Robins, Rotnitzky, and Zhao 1994; Robins and Finkelstein 2000; Hernán, Brumback, and Robins 2000;

Lin 2000; Qi, Wang, and Prentice 2005; Tsiatis 2007; Breslow and Wellner 2007; Saegusa and Wellner 2013). Robins and Finkelstein (2000) applied inverse probability of censoring weights to estimate Cox model that adjusts for dependent censoring by using data collected on time-dependent prognostic factors. IPW has also been applied for Cox models with two-phase stratified sampling under right censoring (Binder 1992; Lin 2000; Breslow and Wellner 2007), while Saegusa and Wellner (2013) further studied the problem of two-phase sampling for Cox model under interval censoring with IPW. Our approach is different from all previous works as we also allow for time-dependent covariates.

Outline. In Section 2, we provide background and notations for our methodological developments on assessing long-term effect through linked dataset. In Section 3, we introduce the NLAC, IPLW, and AIPLW methods and their associated properties. We conduct simulation studies in Section 4 to illustrate the validity of the proposed method. We apply our approach to the linked SWOG PCPT-Medicare study in Section 5. In Section 6, we conclude this article and point out some possible future directions.

2. Long-Term Effect Assessment through Linked Data

We first consider the oracle setting that all participants from the clinical trial are linked. We make a “no gap” assumption such that there is no gap between a participant’s last recorded date within the clinical trial and the start date of the observational follow-up dataset. This “no gap” assumption eliminates the possibility of interval censoring in which a participant is diagnosed with the event of interest while not under observation. For simplicity, we make this “no gap” assumption to focus on the right censoring problem and we discuss how to relax this no gap assumption in Appendix G (supplementary material).

Time is measured since enrollment in the clinical trial. We define T as the failure time, C_1 as the censoring time within the clinical trial and $Q = I(T \leq C_1)$ as the censoring indicator for the clinical trial. We use τ_1 to denote the end time of clinical trial. Possible reasons for censoring in the clinical trial include loss to follow-up and administrative censoring. In contrast, the length of observational follow-up dataset is often determined by the data availability and vary from person to person. We set τ_2 with $\tau_2 > \tau_1$ as the common end time for observational follow-up dataset and assume that there are a significant proportion of participants at risk after τ_2 . Thus, we are interested in estimating the long-term treatment effect on survival up to time τ_2 using data from clinical trial records and observational follow-up. Similarly, we define C_2 as the censoring time in the observational follow-up dataset. Possible reasons for censoring in the observational follow-up include short coverage such that a participant is not covered long enough by the observational dataset, administrative censoring where a participant is event-free and covered by observational follow-up until τ_2 .

We use $C = \max(C_1, C_2)$ to denote the actual censoring time and let $\tilde{T} = \min\{T, C\}$ denote the actual observed time and $\Delta = I(T \leq C)$ be the censoring indicator throughout the entire follow-up period. Let $\mathbf{X} \in \mathbb{R}^p$ denote baseline characteristics, such as clinical factors and treatment assignment. We also use A

to denote the treatment assignment when necessary. We make the independent censoring assumption that T and (C_1, C_2) are conditionally independent given $\mathbf{X}$. Thus, T is conditionally independent of $C = \max(C_1, C_2)$ given $\mathbf{X}$.

2.1. Cox Model and Long-Term Effect

We use Cox model to define the long-term effect and we allow for the possibility of model mis-specification. Assuming that there are n participants in the clinical trial and they are all linked to the observational follow-up dataset. $(\mathbf{X}_i, Q_i, \tilde{T}_i, \Delta_i)$ for $i = 1, \dots, n$ are all observed. To account for potential different treatment effects between clinical trial and the observational follow-up period, we consider the following model with a change point at time τ_1 (Liang, Self, and Liu 1990; Pons 2003)

$$\lambda(t|\mathbf{X}; \beta_0, \beta_1, \theta) = \lambda_0(t) \exp[(\beta_1 + \theta I_{t > \tau_1})A + \beta_0' T \mathbf{X}_{-A}]. \quad (1)$$

β_1 represents the effect of treatment in the clinical trial, while θ represents the difference of the treatment effect between observational follow-ups and clinical trial. $\beta_1 + \theta$ represents the long-term treatment effect. When $\theta = 0$, model (1) reduces to a typical Cox model with no change point. For notational simplicity, we use $\exp(\beta^T \mathbf{X})$ to represent $\exp[(\beta_1 + \theta I_{t > \tau_1})A + \beta_0' T \mathbf{X}_{-A}]$ in the subsequent. Following the notation in Lin and Wei (1989), for the i th individual, let $\lambda_i(t) = \lambda(t|\mathbf{X}_i)$ be the true hazard function, $N_i(t) = I(\tilde{T}_i \leq t, \Delta_i = 1)$ and $Y_i(t) = I(\tilde{T}_i \geq t)$. For $k = 0, 1, 2$, define

$$\mathbf{S}_n^{(k)}(t) = \frac{1}{n} \sum_{i=1}^n Y_i(t) \lambda_i(t) \mathbf{X}_i^{(k)}, \quad \mathbf{s}^{(k)}(t) = \mathbb{E}[\mathbf{S}_n^{(k)}(t)]$$

$$\mathbf{S}_n^{(k)}(\beta, t) = \frac{1}{n} \sum_{i=1}^n Y_i(t) \exp(\beta^T \mathbf{X}_i) \mathbf{X}_i^{(k)},$$

$$\mathbf{s}^{(k)}(\beta, t) = \mathbb{E}[\mathbf{S}_n^{(k)}(\beta, t)]$$

where for a column vector $\mathbf{a}$, denote $\mathbf{a}^{\otimes 2} = \mathbf{a}\mathbf{a}^T$, $\mathbf{a}^{\otimes 1} = \mathbf{a}$ and $\mathbf{a}^{\otimes 0}$ refers to the scalar 1. With Cox model as a working model, the parameter of interest is β_0^* that solves the following equations (Andersen and Gill 1982; Lin and Wei 1989)

$$\begin{aligned} \mathbf{U}_0(\beta) &= \mathbb{E} \left[\Delta \left(\mathbf{X} - \frac{\mathbf{s}^{(1)}(\beta, \tilde{T})}{\mathbf{s}^{(0)}(\beta, \tilde{T})} \right) \right] \\ &= \int_0^{\tau_2} \mathbf{s}^{(1)}(t) dt - \int_0^{\tau_2} \frac{\mathbf{s}^{(1)}(\beta, t)}{\mathbf{s}^{(0)}(\beta, t)} \mathbf{s}^{(0)}(t) dt. \end{aligned} \quad (2)$$

When Cox model is correctly specified with true parameter β_0 , we have $\beta_0^* = \beta_0$. When Cox model is mis-specified, parameter β_0^* that solves (2) is still well-defined and β_0^* can be interpreted meaningfully for many mis-specified Cox models (Lin and Wei 1989).

2.2. The Linkage Problem

Now we consider the more realistic setup that not every participant is linked to the observational follow-up dataset. We use L to denote the linkage status. $L = 1$ means that the participant is linked to the observational follow-up dataset and $L = 0$ means

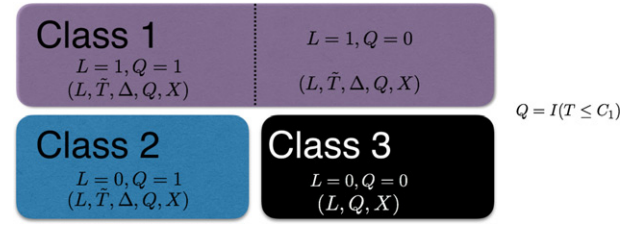


Figure 1. A diagram illustrating the three classes of participants and how it is defined via the linkage indicator L and in-trial censoring indicator Q .

unlinked. We classify participants into three classes based on their linkage status L and in-trial censoring indicator Q .

Class 1: $L = 1$. This class contains all participants linked to the observational follow-up dataset, where we observe the survival outcome: $(L, \tilde{T}, \Delta, Q, X)$.

Class 2: $L = 0, Q = 1$. This class represents participants diagnosed with the event of interest within the clinical trial but unlinked to observational follow-up. We still have complete information $(L, \tilde{T}, \Delta, Q, X)$.

Class 3: $L = 0, Q = 0$. This class includes participants who did not experience the event of interest during the clinical trial, and were not linked to observational follow-up dataset. C_2 is missing and thus C is missing, which causes the missingness of $\tilde{T}$ and Δ . We only observe (L, Q, X) .

Figure 1 provides a summary of the three classes defined above. We have completely observed data $(L, \tilde{T}, \Delta, Q, X)$ in class 1 and 2, while survival outcomes $\tilde{T}, \Delta$ are missing in class 3. To deal with the missing $\tilde{T}$ and Δ , we consider two possible approaches.

3. Methods

3.1. Non-Linked-As-Censored (NLAC) Approach

3.1.1. The Assumption and the NLAC Approach

The first approach, termed *non-linked-as-censored* (NLAC), treats participants from Class 3 ($L = 0, Q = 0$) as censored and sets their censoring time as $C = C_1$. Then we can fit the Cox regression with all participants. For participants from Class 3, NLAC is a natural idea that simply uses their censoring time in clinical trial C_1 as the censoring time for the entire follow-up period. We can view the censoring time as

$$C_{\text{NLAC}} = L \max(C_1, C_2) + (1-L)C_1 = L[\max(C_1, C_2) - C_1] + C_1$$

and NLAC works when $C_{\text{NLAC}} \perp\!\!\!\perp T|\mathbf{X}$. We now present one sufficient linkage assumption for NLAC to work when Cox model is correctly specified.

Theorem 3.1. When Cox model is correctly specified and $(C_1, C_2) \perp\!\!\!\perp T|\mathbf{X}$, if the linkage satisfies the following conditional independent assumption:

$$(A1) \quad L \perp\!\!\!\perp T|\mathbf{X}, C_1, C_2,$$

NLAC provides consistent and asymptotically normal estimates for β_0 .

The proof of Theorem 3.1 can be found in Appendix A. On the other hand, when Cox model is misspecified, the population

parameter β_0^* will depend on the actual distribution of the censoring time $C = \max(C_1, C_2)$. Thus, NLAC leads to biased estimates of β_0^* as the distribution of the censoring time is modified. See Section 3.1.2 for a toy example.

3.1.2. Alternative Approaches and a Motivating Example

There are two alternative approaches that practitioners may use, which are

- Complete-case (CC) analysis that only includes participants that are linked. These corresponds to participants in Class 1 with $L = 1$.
- Complete-case analysis plus (CC+) that includes not only participants that are linked, but also participants with $Q = 1$. This corresponds to participants in Class 1 and 2 in Figure 1. These are the participants with $L + Q > 0$.

We simulate data according to a Cox model with hazard function specified by covariates X_1, X_2, X_3^2 and a Cox model with covariates X_1, X_2, X_3 is fitted. More details are given in Appendix H.5 (supplementary material). Figure 2 presents the 95% confidence intervals for one of the parameters. Oracle method refers to the approach that all participants in the clinical trial are linked. Among all four approaches, CC+ gives the most biased estimates. CC gives less biased estimates than CC+. As anticipated, NLAC also leads to biased estimates. Indeed, in order for CC to be consistent, one sufficient condition is linking completely at random (LCAR), that is, $L \perp (\tilde{T}, \Delta)$. This is similar to the missing completely at random (MCAR) (Little and Rubin 2019) condition. The LCAR condition is a strong condition and may contradict the data.² For CC+, the missingness of survival outcomes now also depends on survival outcomes itself as a participant would be included if $L + Q > 0$. Thus, the missingness can be viewed as MNAR and CC+ will always lead to biased estimates.

3.2. Inverse Probability of Linkage Weighted (IPLW) Approach

3.2.1. The Conditional Linking at Random (CLAR) Assumption

To gain robustness against model mis-specification, we propose the second approach, where different from NLAC as for participants in Class 3, we treat their survival outcomes as missing. Particularly, we assume the following *conditional linking at random (CLAR)* assumption:

$$(A2) L \perp (\tilde{T}, \Delta) | \mathbf{X}, Q = 0$$

More specifically, CLAR states that for a participant that is censored in the clinical trial, their survival outcomes are independent of the linkage status given their baseline covariates, clinical factors and treatment assignment. Note that (A1) actually implies that

$$(A1)' L \perp (\tilde{T}, \Delta) | \mathbf{X}, C_1, C_2.$$

So roughly speaking, the main difference between (A1) and CLAR is that we are replacing C_1 and C_2 by $Q = 0$.

²we can easily test this condition

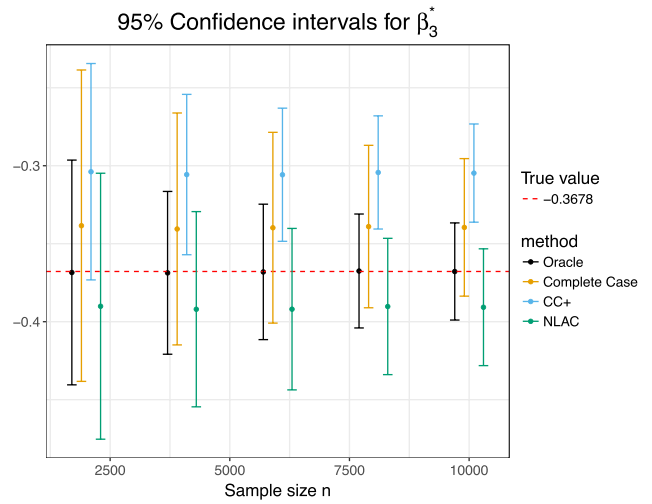


Figure 2. CC+, CC, and NLAC all give inconsistent estimates of β_3^* .

Remark 3.2. We briefly compare CLAR assumption (A2) with the classical MAR type assumption (Rubin 1976; Little and Rubin 2019), which can be written as follows

$$L \perp (\tilde{T}, \Delta) | \mathbf{X}, Q. \quad (3)$$

CLAR is actually implied by MAR (3). However, CLAR is restricting the conditional independence to the subpopulation with $Q = 0$, while MAR (3) is assuming the conditional independence for the whole population. To see why this is important, note that MAR (3) also implies that

$$L \perp (\tilde{T}, \Delta) | Q = 1, \mathbf{X}. \quad (4)$$

Interestingly when $Q = 1$, both $\tilde{T}$ and Δ are observed, meaning that (4) might in fact contradict the data. In contrast, with no assumptions for participants with $Q = 1$, CLAR is non-parametrically identifiable, that is, they will never contradict the data (Robins, Rotnitzky, and Scharfstein 2000). Further, CLAR allows the use of more data compared to MAR. Specifically, an IPW estimator under the MAR assumption would only involve participants that are linked, thereby excluding participants from Class 2. Further discussions on potential linkage assumptions are given in Appendix E (supplementary material).

3.2.2. Inverse Probability of Linkage Weighted (IPLW) Estimator

Since we view $\tilde{T}$ and Δ as missing for participants in Class 3, we can not use the classical partial likelihood for Cox model to estimate the parameters. We first illustrate our approach with time-independent covariates only. We start by writing the regular log-likelihood for Cox model as follows:

$$\begin{aligned} l_n(\beta, \lambda_0) &= \frac{1}{n} \sum_{i=1}^n l(\tilde{T}_i, \Delta_i, \mathbf{X}_i; \beta, \lambda_0) \\ &= \frac{1}{n} \sum_{i=1}^n \log(\lambda(\tilde{T}_i | \mathbf{X}_i)^{\Delta_i} S(\tilde{T}_i | \mathbf{X}_i)) \end{aligned}$$

where $S(t | \mathbf{x})$ is the conditional survival function for the failure time T . However, $l_n(\beta, \lambda_0)$ is unidentifiable since we do not

observe $(\tilde{T}_i, \Delta_i)$ for participants in Class 3 of Figure 1. To resolve the identifiability issue, consider the expected log-likelihood

$$\mathbb{E}(l(\boldsymbol{\beta}, \lambda_0)) = \mathbb{E}[\log(\lambda(\tilde{T}|\mathbf{X})^\Delta S(\tilde{T}|\mathbf{X}))]$$

where $\mathbb{E}$ is the expectation with respect to random variable $(\tilde{T}, \Delta, \mathbf{X})$ and $l(\boldsymbol{\beta}, \lambda_0) = l(\tilde{T}, \Delta, \mathbf{X}; \boldsymbol{\beta}, \lambda_0)$. By the law of large number, we have $l_n(\boldsymbol{\beta}, \lambda_0) \rightarrow_p \mathbb{E}(l(\boldsymbol{\beta}, \lambda_0))$.

Proposition 3.3. Under CLAR assumption, we have

$$\mathbb{E}(l(\boldsymbol{\beta}, \lambda_0)) = \mathbb{E}\left[\frac{I(L + Q > 0)l(\boldsymbol{\beta}, \lambda_0)}{Q + (1 - Q)P(L = 1|\mathbf{X}, Q = 0)}\right] \quad (5)$$

Proposition 3.3 shows that $\mathbb{E}(l(\boldsymbol{\beta}, \lambda_0))$ can be expressed in the inverse probability of linkage weighted (IPLW) form and the proof can be found in Appendix D (supplementary material). We assume a logistic regression model for the linkage probability $P(L = 1|\mathbf{X}, Q = 0)$ for simplicity such that

$$P(L = 1|\mathbf{X}, Q = 0; \boldsymbol{\gamma}_0) = \frac{\exp(\boldsymbol{\gamma}_0^T \tilde{\mathbf{X}})}{1 + \exp(\boldsymbol{\gamma}_0^T \tilde{\mathbf{X}})} = \pi_{\boldsymbol{\gamma}_0}(\mathbf{X})$$

with $\boldsymbol{\gamma}_0 \in \mathbb{R}^{p+1}$ and $\tilde{\mathbf{X}} = (1, \mathbf{X}^T)^T$. In particular, $\boldsymbol{\gamma}_0$ can be estimated by the maximum likelihood estimator $\hat{\boldsymbol{\gamma}}_n$. Using result (5), an IPLW estimator of $\mathbb{E}(l(\boldsymbol{\beta}, \lambda_0))$ is

$$l_n(\boldsymbol{\beta}, \lambda_0) = \frac{1}{n} \sum_{i=1}^n \frac{I(L_i + Q_i > 0)}{Q_i + (1 - Q_i)\pi_{\hat{\boldsymbol{\gamma}}_n}(\mathbf{X}_i)} \times [\Delta_i \log \lambda(\tilde{T}_i|\mathbf{X}_i) + \log S(\tilde{T}_i|\mathbf{X}_i)]$$

with the log-likelihood being weighted by $\hat{w}_i = I(L_i + Q_i > 0)/[Q_i + (1 - Q_i)\pi_{\hat{\boldsymbol{\gamma}}_n}(\mathbf{X}_i)]$. IPLW partial score can then be derived as

$$\hat{\mathbf{U}}_n(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \Delta_i \hat{w}_i \left(\mathbf{X}_i - \frac{\sum_{j=1}^n \hat{w}_j I(\tilde{T}_j \geq \tilde{T}_i) \exp(\mathbf{X}_j^T \boldsymbol{\beta}) \mathbf{X}_j}{\sum_{j=1}^n \hat{w}_j I(\tilde{T}_j \geq \tilde{T}_i) \exp(\mathbf{X}_j^T \boldsymbol{\beta})} \right), \quad (6)$$

which is a sample analog of (2) and $\hat{\boldsymbol{\beta}}_n$ can be obtained by solving (6) using standard statistical software. Detailed derivations of (6) can be found in Appendix B (supplementary material). In summary, our method for estimating the regression parameters of Cox model consists of two steps:

- **Step 1.** We estimate the linkage probability $P(L = 1|Q = 0, \mathbf{X}; \boldsymbol{\gamma}_0) = \pi_{\boldsymbol{\gamma}_0}(\mathbf{X})$ with logistic regression. $\hat{\boldsymbol{\gamma}}_n$ can be obtained by the maximum likelihood estimation.
- **Step 2.** An individual with $\Delta_i = 1$ is weighted with weight $\hat{w}_i$ using the estimated linkage probability $\pi_{\hat{\boldsymbol{\gamma}}_n}(\mathbf{X}_i)$. More specifically, for participants with $Q_i = 1$, the weight is 1; for participants with $Q_i = 0$ and $L_i = 1$, the weight is $1/\pi_{\hat{\boldsymbol{\gamma}}_n}(\mathbf{X}_i)$. $\hat{\boldsymbol{\beta}}_n$ is then obtained by solving (6).

Remark 3.4. It is possible that the CLAR assumption may not be entirely accurate or only holds approximately. A sensitivity analysis (Little et al. 2012) can be conducted to examine how the estimate changes when we slightly perturb the underlying missing data assumption.

Particularly, we can perform a sensitivity analysis to assess the effect of violation of the CLAR assumption using the idea

of exponential tilting (Kim and Yu 2011; Shao and Wang 2016; Zhao et al. 2017). We model the linkage probability as

$$\pi_{\boldsymbol{\gamma}_0, \alpha}(\mathbf{X}, \tilde{T}) = \frac{\exp(\boldsymbol{\gamma}_0^T \mathbf{X} + \alpha \tilde{T})}{1 + \exp(\boldsymbol{\gamma}_0^T \mathbf{X} + \alpha \tilde{T})}, \quad (7)$$

where α is the sensitivity parameter that has to be specified by an analyst. The parameter α may be interpreted as the effect of observed time $\tilde{T}$ on the linkage probability. Equation (7) suggests that the value of α is the effect of one unit of time (in our case, a year) on the log-odds of linkage probability. The parameter $\alpha = 0$ corresponds to the situation when the CLAR assumption is satisfied. A positive α refers to the assumption that individual with a longer observed time has a higher chance to be linked while a negative α is the opposite—the chance of linkage is lower if the observed time is higher. A larger value of $|\alpha|$ leads to a more severe violation of the CLAR.

We begin by selecting a sensitivity parameter vector α 's first. Subsequently, for each α , using the linkage model (7), we can compute the modified weights attributed to α for use in (6), along with the corresponding parameter estimates. Finally, we can evaluate the resulting estimates over the range of α values, representing various degrees of violation from the CLAR assumption.

3.2.3. Time-Dependent Covariates

In practice, it is common for Cox regression to include time-dependent covariates and we now extend our IPLW method to incorporate the time-dependent covariates. We build on the work in Lin and Wei (1989) to extend our theoretical results to include time-dependent covariates with the IPLW partial likelihood estimator. Let $\mathbf{X}_i(t) = (\mathbf{Z}_{1i}^T, \mathbf{Z}_{2i}(t)^T)^T \in \mathbb{R}^p$ denotes the covariates vector, where $\mathbf{Z}_{1i} \in \mathbb{R}^{d_1}$ corresponds to the baseline (time-independent) covariates and $\mathbf{Z}_{2i}(t) \in \mathbb{R}^{d_2}$ corresponds to the time-dependent covariates for $i = 1, \dots, n$ at time t . We assume that $\mathbf{X}(t)$ is observable for all time points and individuals. We have $d_1 + d_2 = p$. $\mathbf{Z}_{2i}(t)$ can represent covariates that are continuously monitored during the clinical trial and observational follow-up datasets. For Cox model with a change point (1), $\mathbf{Z}_{2i}(t) = I(t > \tau_1)\mathbf{A}$. Let $\bar{\mathbf{X}}(t) = \{\mathbf{X}(s) : s \in [0, t]\}$ denotes the history of covariates vector $\mathbf{X}(s)$, up to time t . To incorporate time-dependent covariates into the IPLW partial likelihood, we modify the CLAR assumption (A1) as following:

(D1) The linkage status satisfies that

$$P(L = 1|\tilde{T}, \Delta, Q = 0, \bar{\mathbf{X}}(\tau_M)) = P(L = 1|Q = 0, \mathbf{Z}_1).$$

The distribution of $\mathbf{Z}_1$ is not concentrated on a $(d_1 - 1)$ dimensional affine subspace of $\mathbb{R}^{d_1}$.

(D1) assumes that linkage only depends on time-independent covariates $\mathbf{Z}_1$ and also ensures the identifiability of $\boldsymbol{\gamma}_0$ (see, e.g., Example 5.40 of Van der Vaart 2000). We can further relax this assumption such that linkage also depends on $\mathbf{Z}_2(C_1)$, the value of time-dependent covariates at the censoring time in clinical trial. For simplicity, we assume that linkage only depends on the baseline (time-independent) covariates. Based on assumption (D1), we modify the weights as follows:

$$w_i = \frac{I(L_i + Q_i > 0)}{Q_i + (1 - Q_i)\pi_{\gamma_0}(\mathbf{Z}_{1i})} \quad \widehat{w}_i = \frac{I(L_i + Q_i > 0)}{Q_i + (1 - Q_i)\pi_{\widehat{\gamma}_n}(\mathbf{Z}_{1i})}$$

with $\gamma_0 \in \mathbb{R}^{d+1}$. The IPLW partial score incorporating time-dependent covariates is now as follows:

$$\begin{aligned} \widehat{U}_n(\boldsymbol{\beta}) &= \frac{1}{n} \sum_{i=1}^n \Delta_i \widehat{w}_i \left\{ \mathbf{X}_i(\widetilde{T}_i) - \frac{\mathbf{S}_{n,w}^{(1)}(\boldsymbol{\beta}, \widetilde{T}_i)}{\mathbf{S}_{n,w}^{(0)}(\boldsymbol{\beta}, \widetilde{T}_i)} \right\} \\ &= \frac{1}{n} \sum_{i=1}^n \widehat{w}_i \int_0^{\tau_2} \mathbf{X}_i(t) dN_i(t) - \int_0^{\tau_2} \frac{\mathbf{S}_{n,w}^{(1)}(\boldsymbol{\beta}, t)}{\mathbf{S}_{n,w}^{(0)}(\boldsymbol{\beta}, t)} d\widetilde{N}(t) \end{aligned} \quad (8)$$

with $\mathbf{S}_{n,w}^{(k)}(\boldsymbol{\beta}, t) = \frac{1}{n} \sum_{i=1}^n \widehat{w}_i Y_i(t) \exp(\boldsymbol{\beta}^T \mathbf{X}_i(t)) \mathbf{X}_i(t)^{\otimes k}$ and $\widetilde{N}(t) = \frac{1}{n} \sum_{i=1}^n \widehat{w}_i N_i(t)$.

We redefine $\mathbf{s}^{(k)}(\boldsymbol{\beta}, t) = \mathbb{E}[Y(t) \exp(\boldsymbol{\beta}^T \mathbf{X}(t)) \mathbf{X}(t)^{\otimes k}]$ and $\mathbf{s}^{(k)}(t) = \mathbb{E}[Y(t) \lambda(t|\widetilde{\mathbf{X}}(t)) \mathbf{X}(t)^{\otimes k}]$ where $\lambda(t|\widetilde{\mathbf{X}}(t))$ is the true hazard function for participants with covariates history $\widetilde{\mathbf{X}}(t)$. The estimated parameter $\widehat{\boldsymbol{\beta}}_n$ solves $\widehat{U}_n(\boldsymbol{\beta}) = \mathbf{0}$ and its population version $\boldsymbol{\beta}_0^*$ solves $U_0(\boldsymbol{\beta}) = \mathbf{0}$ with

$$\begin{aligned} U_0(\boldsymbol{\beta}) &= \mathbb{E} \left[\Delta \left(\mathbf{X}(\widetilde{T}) - \frac{\mathbf{s}^{(1)}(\boldsymbol{\beta}, \widetilde{T})}{\mathbf{s}^{(0)}(\boldsymbol{\beta}, \widetilde{T})} \right) \right] \\ &= \int_0^{\tau_2} \mathbf{s}^{(1)}(t) dt - \int_0^{\tau_2} \frac{\mathbf{s}^{(1)}(\boldsymbol{\beta}, t)}{\mathbf{s}^{(0)}(\boldsymbol{\beta}, t)} \mathbf{s}^{(0)}(t) dt. \end{aligned}$$

Now we are ready to present the consistency and asymptotic normality of $\widehat{\boldsymbol{\beta}}_n$.

Theorem 3.5 (Asymptotic results of $\widehat{\boldsymbol{\beta}}_n$). Let $\widehat{\boldsymbol{\beta}}_n$ be the solution to the equation $\widehat{U}_n(\boldsymbol{\beta}) = \mathbf{0}$. Under assumptions (D1)–(D5), we have $\widehat{\boldsymbol{\beta}}_n \rightarrow_p \boldsymbol{\beta}_0^*$ and $\sqrt{n}(\widehat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_0^*) \rightarrow_d N(\mathbf{0}, \Sigma_0^{-1} \Sigma_U \Sigma_0^{-1})$ where $\Sigma_0 = -\frac{\partial U_0(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\boldsymbol{\beta}=\boldsymbol{\beta}_0^*}$ and the form of Σ_U can be found in Theorem B.1 (supplementary material).

Additional technical assumptions (D2–D5) are left in Appendix C (supplementary material). Our proof build on the convergence results for the underlying IPLW empirical process and we give the relevant results in Appendix C (supplementary material).

Remark 3.6. There are two approaches to estimate the covariance and construct the associated confidence interval using Theorem 3.5. The first approach is via the bootstrap—we sample with replacement and re-estimate γ and $\boldsymbol{\beta}$. The second approach is to estimate the asymptotic covariance matrix using the sandwich estimator. This involves estimating both Σ_0 and Σ_U . We can estimate Σ_0 via $\widehat{\Sigma}_0 = -\frac{\partial \widehat{U}_n(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}|_{\boldsymbol{\beta}=\widehat{\boldsymbol{\beta}}_n}$. The covariance matrix Σ_U has an explicit form as (see Theorem B.1 in the supplementary material)

$$\begin{aligned} \Sigma_U &= \text{var}[\mathbf{U}(\boldsymbol{\beta}_0^*)] + \mathbb{E} \left[\mathbf{U}(\boldsymbol{\beta}_0^*) \mathbf{U}(\boldsymbol{\beta}_0^*)^T \frac{I(Q=0)[1 - \pi_{\gamma_0}(\mathbf{Z}_1)]}{\pi_{\gamma_0}(\mathbf{Z}_1)} \right] \\ &\quad - \mathbf{Q}_e(\boldsymbol{\beta}_0^*)^T \Sigma_{\gamma_0}^{-1} \mathbf{Q}_e(\boldsymbol{\beta}_0^*), \end{aligned}$$

where $\mathbf{Q}_e(\boldsymbol{\beta}_0^*) = \mathbb{E}[I(Q=0)(1 - \pi_{\gamma_0}(\mathbf{Z}_1))\widetilde{\mathbf{Z}}_1 \mathbf{U}(\boldsymbol{\beta}_0^*)^T]$ and $\Sigma_{\gamma_0} = \mathbb{E}[I(Q=0)\pi_{\gamma_0}(\mathbf{X})(1 - \pi_{\gamma_0}(\mathbf{X}))\widetilde{\mathbf{X}}\widetilde{\mathbf{X}}^T]$. Σ_{γ_0} can be

estimated by the covariance matrix of $\widehat{\gamma}_n$. For the rest quantities, through a similar proof as Proposition 3.3, we can prove that

$$\begin{aligned} \text{var}[\mathbf{U}(\boldsymbol{\beta}_0^*)] + \mathbb{E} \left[\mathbf{U}(\boldsymbol{\beta}_0^*) \mathbf{U}(\boldsymbol{\beta}_0^*)^T \frac{I(Q=0)[1 - \pi_{\gamma_0}(\mathbf{Z}_1)]}{\pi_{\gamma_0}(\mathbf{Z}_1)} \right] \\ = \text{var} \left[\frac{I(L+Q>0)}{Q+(1-Q)\pi_{\gamma_0}(\mathbf{Z}_1)} \mathbf{U}(\boldsymbol{\beta}_0^*) \right] \end{aligned}$$

and

$$\mathbf{Q}_e(\boldsymbol{\beta}_0^*) = \mathbb{E} \left[I(Q=0) \frac{I(L=1)}{\pi_{\gamma_0}(\mathbf{Z}_1)} (1 - \pi_{\gamma_0}(\mathbf{Z}_1)) \widetilde{\mathbf{Z}}_1 \mathbf{U}(\boldsymbol{\beta}_0^*)^T \right]$$

Thus, each quantity above can be estimated empirically by replacing $\boldsymbol{\beta}_0^*$ with $\widehat{\boldsymbol{\beta}}_n$ and γ_0 with $\widehat{\gamma}_n$.

3.3. Augmented Inverse Probability of Linkage Weighted (AIPLW) Approach

We further developed the augmented inverse probability of linkage weighted (AIPLW) estimator for estimating the long-term effect. The augmented IPLW partial score could be written as

$$\begin{aligned} \widehat{U}_{n,AIPLW} &= \frac{1}{n} \sum_{i=1}^n \left\{ I(Q_i=1) \Delta_i \left[\mathbf{X}_i - \frac{\mathbf{S}_{n,DR}^{(1)}(\boldsymbol{\beta}, \widetilde{T}_i)}{\mathbf{S}_{n,DR}^{(0)}(\boldsymbol{\beta}, \widetilde{T}_i)} \right] \right. \\ &\quad + I(Q_i=0) \left[\frac{I(L_i=1)}{\pi_{\widehat{\gamma}_n}(\mathbf{X}_i)} \Delta_i \left[\mathbf{X}_i - \frac{\mathbf{S}_{n,DR}^{(1)}(\boldsymbol{\beta}, \widetilde{T}_i)}{\mathbf{S}_{n,DR}^{(0)}(\boldsymbol{\beta}, \widetilde{T}_i)} \right] \right. \\ &\quad \left. \left. + \left(1 - \frac{I(L_i=1)}{\pi_{\widehat{\gamma}_n}(\mathbf{X}_i)} \right) \widehat{m}_0(\mathbf{X}_i) \right] \right\}. \end{aligned} \quad (9)$$

The actual forms of $\widehat{m}_0(\mathbf{x})$ and $\mathbf{S}_{n,DR}^{(k)}(\boldsymbol{\beta}, t)$ can be found in Appendix F (supplementary material). On a high level, three regression functions $m_k(\mathbf{x})$ for $k = 1, 2, 3$ need to be estimated to obtain $\widehat{m}_0(\mathbf{x})$ and $\mathbf{S}_{n,DR}^{(k)}(\boldsymbol{\beta}, t)$. We denote these regression function estimators as $\widehat{m}_k(\mathbf{x})$.

Let $\widehat{\boldsymbol{\beta}}_{AIPLW}$ be the solution to $U_{n,AIPLW} = \mathbf{0}$. $\widehat{\boldsymbol{\beta}}_{AIPLW}$ is a doubly robust estimator in the sense that it is a consistent estimator for $\boldsymbol{\beta}_0^*$ when either the linkage probability estimator $\pi_{\widehat{\gamma}_n}(\mathbf{x})$ is consistent for $\pi_0(\mathbf{x})$ or the regression function estimators $\widehat{m}_k(\mathbf{x})$ are consistent for $m_k(\mathbf{x})$ for $k = 1, 2, 3$. However, one limitation is that we need to consistently estimate three regression functions. These three regression functions are themselves variational dependent and congenial parametric modeling can be difficult for all three functions. More details on the illustration for AIPLW estimator could be found in Appendix F.

4. Simulation

We now compare the performances of the proposed NLAC and IPLW methods with several other methods, including complete-case analysis (CC), complete-case analysis plus (CC+) and the oracle method. We consider the following data generation settings. The hazard function is $\lambda(t|\mathbf{X}(t)) = \lambda_0(t) \exp(\beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3(t) \times X_1)$ and $\boldsymbol{\beta}_0 = (\beta_1, \beta_2, \beta_3) = (-\ln(4), \ln(1.5), 0.5)$. X_1 follows a Bernoulli distribution with probability 0.5, X_2 follows a normal distribution with mean and variance both being 1 and $X_3 = I(t \geq \tau_1)$ where τ_1 is the end time of clinical trial. If we treat X_1 as the variable

for treatment assignment, β_3 now represents the difference between the treatment effect after and before τ_1 . This is the Cox model with a change point at τ_1 (1). The baseline hazard function is $\lambda_0(t) = 0.06$. C_1 is exponentially distributed with rate $0.01X_1 + 0.03$ and the censoring time C_2 is set as C_1 plus an exponential random variable with rate $0.05X_1 + 0.03$. Further, we set $\tau_1 = 5$ and $\tau_2 = 16$. Three linkage mechanisms are considered as follows:

1. $P(L = 1) = 0.5$ and L is independent of all other variables. This is the linking completely at random (LCAR) case.
2. $\log \{P(L = 1|X, Q = 0)/P(L = 0|X, Q = 0)\} = -0.25 + 0.5X_1 + 0.5X_2$ and $P(L = 1|Q = 1) = 0.5$. Thus the linkage satisfies CLAR assumption.
3. $\log \{P(L = 1|X, Q = 0, \tilde{T}, \Delta)/P(L = 0|X, Q = 0, \tilde{T}, \Delta)\} = -0.25 + 0.5X_1 + 0.5X_2 - 0.01\tilde{T} - 0.01\Delta$ and $P(L = 1|Q = 1) = 0.5$. This is the linking not at random (LNAR($\tilde{T}$)) case.

Both mechanisms 1 and 2 satisfy our CLAR assumption. Mechanism 2 shows that under the CLAR assumption, data linkage can still depend on the survival outcomes through the in-trial censoring indicator Q . Mechanism 3 slightly violates the CLAR assumption and serves as a case for sensitivity analysis.

We fit two Cox regressions. The first Cox regression is fitted with covariates $X_1, X_2, X_3(t) \times X_1$ and the second Cox regression is fitted with X_1 and X_2 only. Thus, Cox regression is correctly specified for the first regression and mis-specified for the second regression. This suggests that $\beta_0^* = \beta_0$ for the first Cox regression. For the misspecified case, β_0^* is estimated with the oracle method by computing the averages of 1000 parameter estimates with sample size $n = 10,000$. The mis-specified regression omits the time-dependent covariate $X_3(t) \times X_1$ and thus ignores the change-point at τ_1 for the treatment effect. The corresponding parameter β_1^* for X_1 can be interpreted as an averaging effect for the entire follow-up period and is equal to -1.10 , between

the clinical trial treatment effect $-\ln(4) \approx -1.39$ and the observational follow-up treatment effect $-\ln(4) + 0.5 \approx -0.89$.

We use the robust variance estimate (Lin and Wei 1989) when Cox model is misspecified for all methods other than the IPLW method. For the IPLW method, the variance estimate is automatically robust when Cox model is misspecified. When Cox model is correctly specified, one additional mechanism for linkage is considered as

4. $\log \{P(L = 1|X, Q = 0, C_2, \Delta)/P(L = 0|X, Q = 0, C_2, \Delta)\} = -0.25 + 0.5X_1 + 0.5X_2 - 0.1C_2 - 0.1\Delta$ and $P(L = 1|Q = 1) = 0.5$. We call this linkage mechanism LNAR(C_2).

Mechanism 4 is a more serious violation of the CLAR assumption and linkage now depends on the unobserved censoring time C_2 . For all four linkage mechanisms, we use a logistic regression with covariates X_1 and X_2 to estimate the linkage probability $P(L = 1|X, Q = 0)$. The percentages of samples that are not linked and censored in the clinical trial are approximately 39%, 30%, 32%, 51% for these four mechanisms.

Simulation results are reported in Tables 1–2 under sample size $n = 2000$ and we generate 1000 repetitions for each simulation setting. In the table, bias is the difference of the average of 1000 parameter estimates and the true parameter value. Mean standard error (SE) is the average of 1000 SE estimates. CI stands for confidence interval. We first discuss the results when Cox model is correctly specified. When linkage satisfies LCAR, all methods give consistent estimates and correct coverages for the 95% confidence intervals except CC+. Oracle method gives the smallest variance estimates as each participant is linked. NLAC gives the second smallest variance estimates. CC can be viewed as an IPW method with known probability as the weights and it has the largest variance estimates among all methods. The IPLW method gives smaller variance estimates than CC for two reasons. First, IPLW method uses more data than CC; second, IPLW method uses estimated weights, which is known to be

Table 1. Simulation results when Cox model is correctly specified when $n = 2000$.

Mechanism	Method	Bias (Mean SE)			Coverage of 95% CI		
		β_1	β_2	β_3	β_1	β_2	β_3
LCAR	Oracle	−0.00 (0.108)	0.00 (0.033)	−0.00 (0.140)	0.94	0.95	0.96
	CC	−0.00 (0.153)	0.00 (0.046)	−0.01 (0.199)	0.95	0.96	0.96
	CC+	0.13 (0.110)	−0.03 (0.038)	−0.13 (0.168)	0.77 [†]	0.86 [†]	0.89 [†]
	NLAC	−0.00 (0.110)	0.00 (0.038)	−0.01 (0.168)	0.94	0.95	0.95
	IPLW	−0.00 (0.114)	0.00 (0.041)	−0.01 (0.170)	0.94	0.95	0.95
CLAR	Oracle	−0.00 (0.108)	0.00 (0.033)	−0.00 (0.140)	0.94	0.95	0.96
	CC	−0.18 (0.151)	−0.06 (0.045)	0.19 (0.188)	0.79 [†]	0.75 [†]	0.83 [†]
	CC+	−0.07 (0.109)	−0.10 (0.038)	0.09 (0.157)	0.91	0.23 [†]	0.91
	NLAC	−0.00 (0.109)	0.00 (0.037)	−0.00 (0.157)	0.94	0.95	0.95
	IPLW	−0.00 (0.110)	0.00 (0.040)	−0.00 (0.159)	0.94	0.95	0.96
LNAR ($\tilde{T}$)	Oracle	−0.00 (0.108)	0.00 (0.033)	−0.00 (0.140)	0.94	0.95	0.96
	CC	−0.19 (0.151)	−0.07 (0.045)	0.20 (0.189)	0.79 [†]	0.69 [†]	0.84 [†]
	CC+	0.07 (0.109)	−0.11 (0.038)	0.09 (0.158)	0.92	0.16 [†]	0.92
	NLAC	−0.00 (0.109)	0.00 (0.037)	−0.01 (0.158)	0.94	0.95	0.95
	IPLW	−0.00 (0.111)	−0.00 (0.041)	−0.01 (0.160)	0.94	0.95	0.95
LNAR (C_2)	Oracle	−0.00 (0.108)	0.00 (0.033)	−0.00 (0.140)	0.94	0.95	0.96
	CC	−0.31 (0.154)	−0.16 (0.055)	0.33 (0.229)	0.47 [†]	0.16 [†]	0.70 [†]
	CC+	−0.12 (0.111)	−0.22 (0.043)	0.14 (0.202)	0.83 [†]	0.00 [†]	0.88 [†]
	NLAC	−0.00 (0.110)	0.00 (0.042)	−0.02 (0.202)	0.93	0.95	0.95
	IPLW	−0.00 (0.117)	0.01 (0.057)	−0.02 (0.211)	0.93	0.94	0.94

NOTE: We use [†] to highlight settings with coverage below 90%.

Table 2. Simulation results for when Cox model is misspecified and $n = 2000$.

Mechanism	Method	Bias (Mean SE)		Coverage of 95% CI	
		β_1	β_2	β_1	β_2
LCAR	Oracle	−0.00 (0.068)	0.00 (0.033)	0.93	0.94
	CC	−0.00 (0.096)	0.00 (0.046)	0.93	0.96
	CC+	−0.00 (0.081)	−0.03 (0.038)	0.93	0.84 [†]
	NLAC	−0.08 (0.081)	0.00 (0.038)	0.82 [†]	0.95
	IPLW	−0.00 (0.087)	0.00 (0.042)	0.94	0.95
CLAR	Oracle	−0.00 (0.068)	0.00 (0.033)	0.93	0.94
	CC	−0.04 (0.087)	−0.06 (0.045)	0.92	0.76 [†]
	CC+	−0.08 (0.075)	−0.10 (0.038)	0.81 [†]	0.25 [†]
	NLAC	−0.05 (0.076)	0.00 (0.037)	0.89 [†]	0.95
	IPLW	−0.00 (0.079)	0.00 (0.040)	0.93	0.95
LNAR($\tilde{T}$)	Oracle	−0.00 (0.068)	0.00 (0.033)	0.93	0.94
	CC	−0.05 (0.088)	−0.07 (0.046)	0.91	0.70 [†]
	CC+	−0.09 (0.076)	−0.11 (0.038)	0.78 [†]	0.17 [†]
	NLAC	−0.06 (0.077)	−0.00 (0.038)	0.88 [†]	0.94
	IPLW	−0.00 (0.080)	−0.00 (0.041)	0.93	0.95

NOTE: We use [†] to highlight settings with coverage below 90%.

more efficient than IPW method with known probability as weights.

When linkage satisfies CLAR but not LCAR, only oracle method, NLAC and IPLW method give consistent estimate. When CLAR is slightly violated, CC, CC+ all obtain severely biased estimates and confidence intervals with less than nominal coverages. NLAC and IPLW methods still perform relatively well in this case. When linkage depends on the censoring time in observational follow-up C_2 , NLAC gives consistent estimates and correct coverage as expected. For this particular simulation setting, the proposed IPLW method also works pretty well.

Next, we discuss the simulation results when Cox model is misspecified. When linkage satisfies LCAR, CC+, and NLAC give inconsistent estimates of the parameters and do not achieve nominal coverage for 95% confidence intervals. It is expected that CC would perform well in this case as discussed in Section 3.1.2. The IPLW method again obtains smaller variance estimates than CC as more data are fitted and estimated weights improve efficiency. When the linkage satisfies CLAR but not LCAR, only oracle method and IPLW method give consistent estimates and correct coverages. When CLAR is slightly violated, IPLW approach performs the best among all methods other than the oracle method.

We also implement the AIPLW estimator and the simulation results are included in Appendices H.3 and H.4 in the supplementary materials. The results showed that the AIPLW estimator provides unbiased estimates and confidence intervals with nominal coverages when the linkage probability is consistently estimated. When the model for the linkage probability is incorrect, it exhibits competitive performance, even if some of the regression functions could be slightly misspecified.

5. The Linked PCPT-Medicare Claims Study

We apply the proposed NLAC and IPLW method to the SWOG study that links Medicare claims data to the PCPT data (Unger et al. 2018). The PCPT randomly assigned 18,882 eligible men from 1993 to 1997 to finasteride or placebo daily for seven years. PCPT clinical records are linked to participants' Medicare claims data according to common social security

Table 3. SWOG study long-term effect estimation with two change points.

Methods	Finasteride (0–6.5 years)	Finasteride (6.5–7.5 years)	Finasteride (7.5 years +)
IPLW	0.747 (0.661–0.845)	0.702 (0.632–0.779)	1.064 (0.990–1.144)
CC	0.741 (0.648–0.848)	0.700 (0.624–0.784)	1.047 (0.976–1.122)
CC+	0.748 (0.661–0.845)	0.696 (0.628–0.772)	1.045 (0.975–1.120)
NLAC	0.743 (0.657–0.840)	0.696 (0.627–0.773)	1.037 (0.967–1.113)

number, sex and date of birth. Medicare claims are available from 1999 to 2011. The linkage enables PC and death to be identified by both clinical records and Medicare claims. The median time from treatment assignment to the end of the linked dataset was 16 years. We are interested in studying the effect of treatment finasteride on the outcome, which is defined as the time to diagnosis of PC or death, whichever comes first.

We use an “intention to treat” approach in the primary evaluation, where results are analyzed based on participants' initial treatment groups, even if someone did not stick to their assigned treatments. Therefore, it evaluates the effectiveness of the treatment, recognizing that noncompliance is a manifestation of the patient experience with the treatment, and offers insights into their real-world experiences with the treatment. In PCPT, 16.5% of the participants were non-compliant, and the most common reasons were side effect (Pinsky et al. 2021). We recognize that a data linkage of this sort does not resolve issues of noncompliance with an assigned treatment. However, linking our PCPT trial data to Medicare data maintains the advantages of randomization and “intention to treat” approach allows for an unbiased evaluation of the intervention's long-term effectiveness.

Out of the total 18,882 participants, 2168 had a gap between the end of the SWOG trial and the start of Medicare claims, with a median gap length of 1.6 years. We excluded these participants from the analysis. Additionally, participants with missing covariates were removed, leaving 16,518 participants (finasteride = 8253; placebo = 8265) in the final analysis. Of these, 11,858 (71.8%) participants were linked to Medicare claims, falling under Case 1. Among this group, 10,041 participants were censored during the trial but were later linked to Medicare claims, demonstrating that the linkage provided additional follow-up information for the majority of participants. Another 411 participants (2.5%) experienced the event of interest during the clinical trial but were not linked to observational follow-up (Case 2). Finally, 4249 participants (25.7%) were censored during the trial and did not have a subsequent link to Medicare claims (Case 3).

We fit Cox regression with covariates including the prostate-specific antigen (PSA) level at study entry, race, body mass index at study entry, first degree family history of prostate cancer, age at baseline. Additional covariates were included for logistic regression modeling linkage: participants' education level, marital status, employment status, type of jobs.

The PCPT was designed to identify prostate cancer (PC) in several ways: (a) biopsies triggered by concerns from annual digital rectal exams suggestive of PC; (b) biopsies due to elevated PSA levels; (c) end-of-study biopsies in the seventh year, which were recommended for all participants who had not been previously diagnosed with PC. As a result, a significant number of

PC cases were identified in the 7-year biopsy window (6.5–7.5 years), leading to a noticeable increase in the hazard rate during this period (Unger et al. 2018). Therefore, Cox regressions with two change points at 6.5 and 7.5 years are fitted to accommodate the design. We compared the results of CC, CC+, NLAC, and IPLW method in Table 3. Table 3 contains the parameter estimates and 95% confidence intervals for treatment Finasteride in different time periods.³ Overall, the results do not differ too much between all four methods based on the 95% confidence intervals. A key reason might be that the linkage rate was high for the original study (Unger et al. 2018) as 75% of the participants were linked. Further, they examined potential health care utilization differences by arm and other potential biases in Unger et al. (2018) and found no evidence of strong differences. This suggests that linkages might be following a LCAR mechanism. We also obtained robust variance estimates (Lin and Wei 1989) and the corresponding confidence intervals. The results are very similar to the nonrobust ones. In summary, finasteride arm participants had a 30% decrease in the hazard ratio of prostate cancer (hazard ratio (HR) = 0.75, 95% confidence intervals (CI) = 0.66–0.85) during the first 6.5 years. The effect of finasteride is strongest between 6.5 and 7.5 years (HR = 0.70, 95% CI = 0.63–0.78). The long-term effect of finasteride after the 7.5 years does not seem to increase the risk of PC (HR = 1.06, 95% CI = 0.99–1.14). We further fit a Cox regression without any change points and the results are shown in the supplementary material. The results again does not differ too much between all four methods. In summary, the long-term effect of finasteride, now estimating the averaging effect over the entire follow-up period, is still beneficial (HR = 0.90, 95% CI = 0.85–0.95).

6. Discussion

In this article, we consider the problem of long-term effect estimation by fitting a Cox model to a partially linked dataset. We propose three methods, NLAC, IPLW, and AIPLW, to estimate the underlying long term effect. In summary, NLAC is a natural and easy to implement method that works well when Cox model is correctly specified and linkage satisfies a conditional independent assumption. On the other hand, IPLW and AIPLW are more robust to model mis-specification for Cox model and relies on a different linkage assumption, the CLAR assumption. When confidence exists in a correctly specified Cox model and the conditional independent linkage assumption, we recommend using the NLAC method due to its superior efficiency. However, if there is evidence suggesting that the Cox model may be violated, we recommend to consider IPLW or AIPLW methods. AIPLW has gained popularity with its robustness and efficiency, and our numerical results demonstrate the robustness of AIPLW under model misspecification. However, Kang and Schafer (2007) warn against the use of doubly robust estimators when both working models are misspecified. Further discussions highlight that the choice of nuisance parameter estimates can significantly influence the properties of the doubly robust estimators when at least one working model is misspecified (Tan 2007; Tsiatis and Davidian 2007; Robins et al. 2007). In our case, constructing the AIPLW estimator requires specifying

multiple outcome regression functions. The optimal approach to estimating these nuisance functions remains an open question and is an area for future research. Additionally, we point out a few other potential future directions for this work:

- *Interval Censoring.* We have made the “no gap” assumption in the article to focus on the right censoring problem for simplicity. However, in practice, it is possible that there might be gaps between the clinical trial and observational follow-up dataset. Thus, to fully deal with the problem, we need to extend our current procedure to the interval censoring case as mentioned in Appendix G (supplementary material). Saegusa and Wellner (2013) has studied the problem of two-phase sampling for Cox models under interval censoring. In Zeng, Mao, and Lin (2016), they formulate potentially time-dependent covariates effects on the interval-censored failure time through a broad class of semiparametric transformation models. These models include Cox model as a special case. These models offer potential utility in addressing gap cases within our application. We defer the extension of the proposed approach for interval censored data to future work.
- *Handling change points.* We believe due to the study design, it is reasonable to assume change points are known in our SWOG data application. However, as reviewer points out, in many studies, the choice of change points may not be obvious. We can consider estimating the change-point location (Gijbels and Gürlér 2003). Our proposed method can be extended to accommodate the estimated change point and we leave this as a future work.
- *Competing risks.* With the prolonged follow-up, competing risks could occur to prevent the occurrence of the primary event of interest. For example, in the SWOG-Medicare linkage example, if the primary event or outcome of interest is time to diagnosis of prostate cancer, or other non-cancer outcomes such as ischemic and thrombotic events, we should analyze the cumulative incidence of events accounting for potential competing risk of death. For competing risk data, Fine and Gray (1999) developed a proportional subdistribution hazards model to directly assess the effect of a covariate on the cumulative incidence function. We plan to extend the Fine and Gray model to the trial-linkage studies for evaluating long-term effects, which we will address in future work.

Supplementary Materials

Appendix A contains the proof for Theorem 3.1 and discusses linkage assumptions for NLAC. Appendix B contains the derivation of IPLW partial score. Appendix C and D contains the results for IPLW empirical process theory and the proof for technical results. Appendix E contains more discussions of linkage assumptions. Appendix F discusses the doubly robust AIPLW estimator. Appendix G discusses the relaxation of the “no gap” assumption. Appendix H contains more numerical results. Appendix I includes a discussion on extension of the proposed methods to alternative models.

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³on the exponential level

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The authors report there are no competing interests to declare.

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